

Oil supply shocks and the U.S. economy: An estimated DSGE model[☆]

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ABSTRACT

We develop and use a medium-sized DSGE model of the U.S. economy to evaluate how U.S. real GDP responds to oil price movements that originate from global oil supply shocks. The core of the model is a standard macroeconomic DSGE framework that includes nominal and real frictions. The model includes oil as an input in multiple domestic sectors (consumption, intermediate goods, and transportation services). We include a domestic oil production sector for the United States to reflect the recent development in shale oil technology. The model also captures international trade in goods and oil. The model parameters are set through a combination of calibration and Bayesian estimation using quarterly data for 1991 through 2015. Baseline estimation of the model finds the elasticity of U.S. real GDP with respect to an oil price shock of -0.015 , which is among the less elastic estimates in the literature. Using the model to conduct counterfactual analysis, we find that decreasing steady state U.S. oil consumption substantially reduces the response of real GDP to oil prices. Increasing U.S. domestic oil production only modestly reduces the response of real GDP to oil prices.

1. Introduction

Since the early 1970s, economic research has focused on how the U.S. economy has responded to world oil price shocks. As surveyed by Brown and Yücel (2002), Jones et al. (2004), and Kilian (2008c), the aggregate economic effects were interpreted as the consequences of unfavorable oil supply shocks—with the consequences being rising oil prices, slower real GDP growth (possibly recession), higher unemployment rates, and higher price levels. As oil prices have fluctuated throughout the 2000s, however, they seem to have yielded a much smaller response in U.S. real GDP.

Some of the explanations for a weaker response have included increased global financial integration, greater flexibility of the U.S. economy (including labor and financial markets), the reduced oil intensity of the U.S. economy, increased experience with oil price shocks, better monetary policy, and smaller and less frequent shocks. Contributions include Huntington (2003), Stock and Watson (2003), Nakov and Pescatori (2009), Blanchard and Gali (2010), Edelstein and Kilian (2007), Herrera and Pesavento (2009) and Kilian and Lewis (2011).

A growing literature examines the implications of distinguishing the effects of oil price movements that arise from demand or supply shocks

in the oil market. That is, oil supply and demand shocks can have different implications for aggregate economic activity. The newer efforts include Kilian (2008 a,b, 2009, 2014), Balke et al. (2010), Baumeister and Hamilton (2015), and Baumeister and Kilian (2016). According to Kilian (2008b), Balke et al., and Baumeister and Hamilton, the differing sources of oil price shocks can explain some of the apparent changes in the relationship between oil price movements and aggregate economic activity. Nonetheless, the newer empirical literature, such as Kilian and Vigfusson (2011 a,b), Kilian (2014), Baumeister and Hamilton, and Baumeister and Kilian, suggest that oil price increases arising from supply shocks in the oil market have a much smaller effects on aggregate economic activity than is found in the earlier literature.

Dynamic stochastic general equilibrium (DSGE) modeling is an approach that allows for identification of the various sources of oil price shocks and at the same time can capture alternative explanations of why the economy has become less sensitive to oil price shocks. Although DSGE models are widely used to examine aggregate economic activity and macroeconomic policy, few empirical exercises involving DSGE models have been used to examine the effect of oil supply shocks on world oil prices and aggregate economic activity. Notable exceptions include Bodenstein et al. (2011), Bodenstein and Guerrieri (2011), and Balke et al. (2010). Although these efforts identified various sources of

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oil price shocks, the efforts represented U.S. oil consumption rather broadly, and none of these efforts allowed for the frictions that are now typically found in DSGE models.

In the present analysis, we expand inquiry by using a medium-sized DSGE model of the U.S. economy that also represents the world oil market, U.S. international trade, and aggregate economic activity in the rest of the world (ROW). The model provides a mapping from structural shocks—such as those in technology and preferences—to observables such as oil prices, oil production, and other measures of economic activity. We use a combination of calibration and Bayesian methods to determine the model's parameters and to assess the stochastic process generating the exogenous shocks. The latter allows us to identify oil supply shocks and to estimate their effects on world oil prices and U.S. real GDP.

The core of the model is essentially a standard macroeconomic DSGE framework that includes nominal frictions (prices) and real frictions (real wages and capital and labor adjustment costs). To this model, we introduce oil as an additional input to production (and consumption), multiple sectors (consumption, final goods, intermediate goods, transportation services and oil production). The transportation sector captures an important channel through which oil prices can affect aggregate economic activity. In addition, the domestic (U.S.) economy interacts with the ROW through the world oil market and final goods market, which provides channels through which U.S. economic activity and these markets affect each other.

The model is used to examine how U.S. real GDP responds to oil price movements that are the result of an oil supply shock originating in the ROW. As a summary measure of this effect, we calculate the elasticity of U.S. real GDP with respect to oil price changes due to a ROW oil supply shock. We estimated the mean (median) of the posterior distribution of the GDP/oil price elasticity to be -0.015 (-0.017). That is, if oil prices increase by 10% due to oil supply shock originating in the ROW, U.S. real GDP falls by 0.15% over the subsequent year.

We also use the model to conduct counterfactual analyses. In particular, we examine how a decline in the U.S. oil usage intensity and an increase in U.S. oil production affects long-run oil consumption, oil imports, as well as the response of the U.S. economy to oil supply shocks originating in ROW. We find that if the steady state level of U.S. oil consumptions is lower, the U.S. economy becomes substantially less responsive to ROW oil supply shocks. In the model when steady state U.S. oil production is substantially higher (75% higher), there is only a modest reduction of the sensitivity of U.S. economic activity to ROW oil supply shocks. Reducing U.S. oil consumption appears to have more impact in lowering the sensitivity of aggregate U.S. economic activity than an increase in U.S. oil production.

The remainder of the paper is organized as follows. Section 2 describes the DSGE model of the U.S. economy, the world oil market, U.S. international trade and ROW economic activity. Section 3 covers the estimation of the model and the posterior distribution of some of the key parameters in the model. Section 4 provides the key empirical results of the model through impulse response functions and historical decompositions. Section 5 provides estimated elasticities of real GDP with respect to the oil price. Section 6 provides counterfactual analysis of the model whereby we change the steady state oil usage intensity and domestic oil production and examine their effects on the GDP/oil price elasticity. Section 7 offers concluding remarks and policy implications.

2. Model

Our model builds on the work of open economy models of oil such as Backus and Crucini (2000), Balke et al. (2010), Bodenstein et al. (2011), and Bodenstein and Guerrieri (2011). Where we differ is that we allow a specific role for transportation in the use of oil products and for oil efficiency to evolve endogenously. Unlike these other models, we do not model structurally the rest-of-the world because data limitations make estimating the structural parameters for the rest-of-the-world

difficult.

The main features of the model are as follows:

1. U.S. Economy
 - a. Largely neoclassical, dynamic macroeconomic model with real wage and nominal price rigidities, adjustment costs for capital and labor.
 - b. Multi-sector model with emphasis on oil's impact through transportation sector and through consumption.
 - c. Endogenous oil efficiency.
2. World oil market
 - a. Quasi-structural model of ROW supply and demand for oil
 - b. Interacts with U.S. oil supply and demand in world oil market to determine price of oil
3. Non-oil interaction of U.S. with ROW
 - a. Reduced form model of trade in non-oil goods between U.S. and ROW. Model allows for feedback from U.S. and oil markets to ROW. In the long-run, an increase in U.S. oil imports must be met by an increase in net exports of the non-oil good to ROW, but that need not hold in the short-run.

2.1. Domestic (U.S.) household sector

Household utility is given by:

$$\sum_{i=0}^{\infty} \beta^i u(c_t, l_t) \quad (1)$$

where in the benchmark model

$$u(c_t, l_t) = \frac{(c_t - x_c h_t)^{1-\sigma}}{1-\sigma} - \chi \frac{l_t^{1+\eta}}{1+\eta} \quad (2)$$

where h_t represents habits (external) that evolve according to $h_t = c_{t-1}^{1-\gamma_h} h_{t-1}^{\gamma_h}$. Consumption, c_t , is given by:

$$c_t = z_{c,t} (\psi_n n_t^{\rho_c} + (1-\psi_n) k_{sc,t}^{\rho_c})^{1/\rho_c} \quad (3)$$

and

$$l_t = \left(1 + \phi \left(\frac{l_{f,t}}{l_{f,t-1}} \right) \right) l_{f,t} + \left(1 + \phi \left(\frac{l_{m,t}}{l_{m,t-1}} \right) \right) l_{m,t} + \left(1 + \phi \left(\frac{l_{o,t}}{l_{o,t-1}} \right) \right) l_{o,t} \quad (4)$$

Consumption depends on expenditures on nondurables, n_t , and the service flow from durables, $k_{sc,t}$. Households allocate labor to production of final goods, $l_{f,t}$, intermediate goods, $l_{m,t}$, and oil production, $l_{o,t}$. The $\phi \left(\frac{l_{j,t}}{l_{j,t-1}} \right)$ terms represent adjustment costs to households of re-allocating their labor supply. We use the functional form:

$$\phi_l \left(\frac{l_{j,t}}{l_{j,t-1}} \right) = a_{jl} \frac{(l_{j,t}/l_{j,t-1} - 1)^2}{2} \quad (5)$$

We assume real wage frictions similar to those in Blanchard and Gali (2010). Here the real wage (in terms of the final good) before adding the adjustment costs of changing labor is given by

$$w_t = (w_{t-1})^{\alpha_w} (\lambda_{t,t})^{1-\alpha_w} \quad (6)$$

where

$$\lambda_{t,t} = \left(\frac{1}{\rho_w} \right) \lambda_{c,t} \left(\frac{-U_l(c_t, l_t)}{U_c(c_t, l_t)} \right) \quad (7)$$

$\lambda_{c,t}$ is the real shadow price of the consumption good, $\left(\frac{-U_l(c_t, l_t)}{U_c(c_t, l_t)} \right)$ is the marginal rate of substitution between leisure and consumption, and $\left(\frac{1}{\rho_w} \right)$ is a markup greater than one.

The service flow from durables depends on the stock of durables, $k_{c,t}$, and oil usage, $o_{c,t}$:

$$k_{sc,t} = z_{ksc,t} (\psi_{co}(e_{c,t-1} o_{c,t})^{\rho_{ksc}} + (1-\psi_{co})k_{c,t-1}^{\rho_{ksc}})^{1/\rho_{ksc}} \quad (8)$$

The term $e_{c,t-1}$ represents the current period's oil efficiency in generating the flow of services from consumer durables. Note that the only margin with which to change capital services in the current period is by oil usage ($o_{c,t}$), as oil efficiency and the stock of durables are predetermined in the current period. In this application, we are thinking of oil consumption ($o_{c,t}$) as gasoline consumption by households and the durables as representing automobiles.

The stock of durables evolves according to a standard capital accumulation equation:

$$k_{c,t} = (1-\delta)k_{c,t-1} + z_{I,t} I_{c,t} \quad (9)$$

where $I_{c,t}$ is purchases of new durables. The $z_{I,t}$ represents an investment technology shock which is common to durable and capital goods across the economy. Oil usage efficiency evolves according to:

$$e_{c,t} = v_{c,t} x_{c,t} \delta e_{c,t-1}^{1-\delta} \quad (10)$$

where $x_{c,t}$ is most recent increment to efficiency (relative to $e_{c,t-1}$) and δ is the depreciation rate of durables.

Household budget constraint is given by:

$$(w_{f,t} l_{f,t} + w_{m,t} l_{m,t} + w_{o,t} l_{o,t}) + p_{f,t} + p_{r,t} + p_{ksm,t} + p_{ro,t} - p_{f,t} n_t - p_{f,t} (1 + \phi_I(I_{c,t}/k_{c,t-1}))(1 + \phi_e(x_{c,t}))I_{c,t} + (1+R_t)A_t - A_{t+1} = 0 \quad (11)$$

where $p_{f,t}$, $p_{r,t}$, $p_{ksm,t}$, and $p_{ro,t}$ are profits from firms in the final goods sector, intermediate goods sector, intermediate capital services providers, and oil producers, respectively and A_t is the net assets of domestic households (relative to the ROW). We do not model this interaction with the ROW explicitly but with a reduced form relationship with feedback from U.S. to ROW.

The term $\phi_I(I_{c,t}/k_{c,t-1})$ reflects adjustment costs of changing consumer durables. In the steady state, we assume these adjustment costs have the following properties: $\phi_I(\delta) = 0$, $\phi_I'(\delta) = 0$, and $\phi_I''(\delta) > 0$. The term $\phi_e(x_{c,t})$ captures the cost of changing efficiency from its steady state value. We normalize this cost function so that in the steady state this cost is zero. We assume that in the steady state: $\phi_e(x_{c,ss}) = 0$, $\phi_e'(x_{c,ss}) > 0$, and $\phi_e''(x_{c,ss}) = 0$. We interact the costs of increasing efficiency with those of investment to capture the notion that efficiency improvements are embodied in capital and that increasing efficiency increases investment costs. We use the following functional forms:

$$\phi_I(I_{c,t}/k_{c,t-1}) = a_{ck} \frac{(I_{c,t}/k_{c,t-1} - \delta)^2}{2} \quad (12)$$

$$\phi_e(x_{c,t}) = a_{ec}(x_{c,t} - x_{c,ss}) + b_{ec} \frac{(x_{c,t} - x_{c,ss})^2}{2} \quad (13)$$

2.2. Final goods production

Final goods are used in consumption (nondurables) and in investment in the various types of capital and in consumer durables. Final goods producers take prices as given. Final goods producers take intermediate goods, $y_{m,t}$, and transportation services, $y_{tr,t}$, to produce the final good:

$$y_{f,t} = z_{f,t} (\psi_m y_{m,t}^{\rho_f} + (1-\psi_m) y_{tr,t}^{\rho_f})^{1/\rho_f} \quad (14)$$

The intermediate input is a composite good of a continuum of differentiated goods given by:

$$y_{m,t} = \int_0^1 (y_{m,t}(z))^{\rho_m} dz \quad (15)$$

Demand for the z -th intermediate good is given by:

$$y_{m,t}(z) = \left(\frac{p_{m,t}}{p_{m,t}(z)} \right)^{\frac{1}{1-\rho_m}} y_{m,t} \quad (16)$$

where $p_{m,t}(z)$ is the price of the z th variety and $p_{m,t}$ is a price index over varieties given by:

$$p_{m,t} = \int_0^1 \left(p_{m,t}(z)^{\frac{\rho_m}{\rho_m-1}} \right)^{(\rho_m-1)/\rho_m} dz \quad (17)$$

Transportation input is a function of labor and capital services in transport:

$$y_{tr,t} = z_{tr,t} (\psi_{tr} l_{tr,t}^{\rho_{tr}} + (1-\psi_{tr}) k_{str,t}^{\rho_{tr}})^{1/\rho_{tr}} \quad (18)$$

In turn, capital services in transport is a function of capital and oil usage:

$$k_{str,t} = z_{kstr,t} (\psi_{o,ir}(e_{tr,t-1} o_{tr,t})^{\rho_{kstr}} + (1-\psi_{o,ir})k_{tr,t-1}^{\rho_{kstr}})^{1/\rho_{kstr}} \quad (19)$$

Again, we are thinking of oil usage as primarily gasoline and the capital as equipment used in transportation (trucks, trains, aircraft and associated infrastructure). The size of the transportation sector in the final good is relatively small but it is an important source of oil usage when we calibrate the model to the U.S. economy.

Final goods producers maximize the present value of profits:

$$\sum_{i=0}^{\infty} M_{t,t+i} (p_{f,t+i} y_{f,t+i} - w_{tr,t+i} l_{tr,t+i} - p_{m,t+i} y_{m,t+i} - p_{f,t+i} (1 + \phi_I(I_{tr,t+i}/k_{tr,t+i-1}))(1 + \phi_e(x_{tr,t+i}))I_{tr,t+i}) \quad (20)$$

where $M_{t,t+i}$ is the stochastic discount factor of the representative household and is defined below. Again, the cost of investment depends on $\phi_I(I_{tr,t+i}/k_{tr,t+i-1})$ and $\phi_e(x_{tr,t+i})$ which have similar properties to the function described above for consumption. Capital in transportation services evolves according to:

$$k_{tr,t} = (1-\delta)k_{tr,t-1} + z_{tr,t} I_{tr,t} \quad (21)$$

while efficiency evolves according to:

$$e_{tr,t} = x_{tr,t} \delta e_{tr,t-1}^{1-\delta} \quad (22)$$

where $x_{tr,t}$ is most recent increment to efficiency (relative to $e_{tr,t-1}$).

2.3. Intermediate input production

Intermediate goods producers use labor and capital services to produce a differentiated good. The production technology is given by:

$$y_{m,t}(z) = z_{m,t} l_{m,t}(z)^{\psi_{m,t}} k_{sm,t}(z)^{1-\psi_{m,t}} \quad (23)$$

where $z_{m,t}$ is a common productivity factor.¹ Intermediate good producers face a monopolistically competitive market structure. Since intermediate goods producers are price setters, they face a demand for their product given by:

$$y_{m,t}(z) = \left(\frac{p_{m,t}}{p_{m,t}(z)} \right)^{\frac{1}{1-\rho_m}} y_{m,t} \quad (24)$$

Intermediate goods producers maximize the present value of profits:

$$\sum_{i=0}^{\infty} M_{t,t+i} \left(p_{m,t+i}(z) y_{m,t+i}(z) - w_{m,t+i} l_{m,t+i}(z) - r_{m,t+i} k_{sm,t+i}(z) - p_{f,t+i} \phi_p \left(\frac{p_{m,t+i}(z)}{p_{m,t+i-1}(z)} \right) y_{m,t+i}(z) \right) \quad (25)$$

where $r_{m,t}$ is the rental rate for capital services used by intermediate

¹ We assume Cobb-Douglas for intermediate goods production, in part, to make aggregation of input demands across varieties straightforward.

goods producer $z(k_{sm,t}(z))$. We assume that producers face Rotemberg-type adjustment costs to changing prices which are given by

$$\phi_p \left(\frac{p_{m,t}(z)}{p_{m,t-1}(z)} \right) = a_c \frac{\left(\frac{p_{m,t}(z)}{p_{m,t-1}(z)} \pi_{m,t-1}^{\frac{1}{\epsilon}} - 1 \right)^2}{2}, \text{ where } \epsilon_{\pi} \text{ is an indexing parameter and } \pi_{m,t-1} \text{ is the gross inflation rate for aggregate intermediate goods prices. One can rewrite the intermediate goods producer's problem as}$$

$$\sum_{i=0}^{\infty} M_{t,t+i} \left((p_{m,t+i}(z) - \lambda_{m,t+i} - p_{f,t+i} \phi_p \left(\frac{p_{m,t+i}(z)}{p_{m,t+i-1}(z)} \right)) y_{m,t+i}(z) \right) \quad (26)$$

$\lambda_{m,t}$ is marginal cost of production and is the same across intermediate goods producers:

$$\lambda_{m,t} = z_{m,t}^{-1} \left(\frac{w_{m,t}}{\psi_{m,l}} \right)^{\psi_{m,l}} \left(\frac{r_{m,t}}{1 - \psi_{m,l}} \right)^{1 - \psi_{m,l}} \quad (27)$$

Intermediate goods producers choose $p_{m,t}(z)$ to maximize the present value of profits.

2.4. Capital goods suppliers to intermediate good production

There are suppliers of capital services to intermediate goods producers. The suppliers of capital services to intermediate goods producers choose oil and capital to maximize the present value of profits:

$$\sum_{i=0}^{\infty} M_{t,t+i} (r_{m,t+i} k_{sm,t+i} - p_{o,t+i} o_{m,t+i} - p_{f,t+i} (1 + \phi_f(I_{m,t+i}/k_{m,t+i-1})) (1 + \phi_e(x_{m,t+i})) I_{m,t+i}) \quad (28)$$

Capital services in intermediate goods is given by:

$$k_{sm,t} = z_{sm,t} (\psi_{m,o} (e_{m,t-1} o_{m,t})^{\rho_{ksm}} + (1 - \psi_{m,o}) k_{m,t-1}^{\rho_{ksm}})^{1/\rho_{ksm}}, \quad (29)$$

along with

$$k_{m,t} = (1 - \delta) k_{m,t-1} + z_{f,t} I_{m,t} \quad (30)$$

and

$$e_{m,t} = x_{m,t} \delta e_{m,t-1}^{1-\delta} \quad (31)$$

One can think of $o_{m,t}$ as non-transport use of oil in the production of intermediate goods, similar to the use of oil as an input to petroleum products.

2.5. Domestic oil production

Domestic oil production depends on labor and capital. We abstract from depletion; depletion would be reflected in the depreciation in the capital of oil production sector. Domestic oil producers maximize:

$$\sum_{i=0}^{\infty} M_{t,t+i} (p_{o,t+i} y_{o,t} - w_{o,t+i} l_{o,t+i} - p_{f,t+i} (1 + \phi_f(I_{o,t+i}/k_{o,t+i-1})) I_{o,t+i}) \quad (32)$$

Oil production depends on labor and capital services

$$y_{o,t} = z_{o,t} (\psi_{o,l} (l_{o,t})^{\rho_o} + (1 - \psi_{o,l}) k_{o,t-1}^{\rho_o})^{1/\rho_o} \quad (33)$$

with

$$k_{o,t} = (1 - \delta) k_{o,t-1} + z_{f,t} I_{o,t} \quad (34)$$

2.6. Monetary policy and interest rates

We assume that the U.S. monetary policy follows a Taylor-like rule:

$$R_t = \frac{R_{ss}}{\pi_{ss}^*} \pi_t^* \left(\left(\frac{\pi_{GDP,t}}{\pi_t^*} \right)^{\gamma_{\pi}} \left(\frac{y_{GDP,t}}{y_{GDP,ss}} \right)^{\gamma_y} \right)^{1-\gamma_R} \left(\frac{R_{t-1}}{R_{ss}} \right)^{\gamma_R} \epsilon_{R,t} \quad (35)$$

$\pi_{GDP,t}$ is the gross inflation rate in the GDP deflator, π_t^* is the target inflation rate, and “ss” denotes long-run or steady-state values of the

variables. We include the target inflation rate to help capture the general downward drift in inflation and nominal interest rates over our sample.²

We also allow for “risk” or preference shocks. The standard Euler equation for the “risk free” rate is:

$$E \left[(R_t / \pi_{f,t+1}) \beta \frac{u_c(c_{t+1}, l_{t+1}) / \lambda_{c,t+1}}{u_c(c_t, l_t) / \lambda_{c,t}} \right] = 1 \quad (36)$$

where $\lambda_{c,t}$ is the price of consumption relative to the final goods price. For the discount factor for firm profits, we include an additional term:

$$M_{t,t+1} = z_{risk,t} \beta \frac{u_c(c_{t+1}, l_{t+1}) / \lambda_{c,t+1}}{U_C(c_t, l_t) / \lambda_{c,t}} \quad (37)$$

When the model is log-linearized, $z_{risk,t}$ acts like an exogenous risk premium when discounting future profits. This in turn affects the optimal amount of investment in capital and oil efficiency. This “shock” is meant to capture financial frictions that are not explicitly modeled that affect investment decisions. In our framework, a positive risk shock would result in a decline in the risk premium.

2.7. Market clearing

The domestic (U.S.) economy interacts with the ROW through trade in final goods, oil, and assets. For the final goods market,

$$\begin{aligned} y_{f,t} - n_t - (1 + \phi_f(I_{c,t}/k_{c,t-1}))(1 + \phi_e(x_{c,t})) I_{c,t} \\ - (1 + \phi_f(I_{tr,t}/k_{tr,t-1}))(1 + \phi_e(x_{tr,t})) I_{tr,t} \\ - (1 + \phi_f(I_{m,t}/k_{m,t-1}))(1 + \phi_e(x_{m,t})) I_{m,t} - (1 + \phi_f(I_{o,t}/k_{o,t-1})) I_{o,t} \\ + p_{f,t} \phi_p \left(\frac{p_{m,t}}{p_{m,t-1}} \right) y_{m,t} - nx_{f,t} = 0 \end{aligned} \quad (38)$$

Production of the final good is allocated, in part, among nondurable consumption (n_t) and investment and the adjustment costs associated with investment and efficiency in consumption, the production of transportation services, and intermediate goods $((1 + \phi_f(I_{j,t}/k_{j,t-1}))(1 + \phi_e(x_{j,t})) I_{j,t}$ for $j = c, tr, m$). The term, $p_{f,t} \phi_p \left(\frac{p_{m,t}}{p_{m,t-1}} \right) y_{m,t}$, in Eq. (38) represents resources used in changing intermediate goods prices while $nx_{f,t}$ is net exports of the final good to the ROW.

For the oil market,

$$y_{o,t} - o_{c,t} - o_{tr,t} - o_{m,t} + row_{s,t} - row_{d,t} = 0 \quad (39)$$

where $y_{o,t}$ is domestic oil production, $o_{c,t}$ is oil used in domestic consumption, $o_{tr,t}$ is oil used in domestic transportation services, $o_{m,t}$ is oil used in domestic intermediate goods production while $row_{s,t}$ and $row_{d,t}$ are ROW oil supply and demand, respectively. Note that net oil imports into the U.S. is given by: $o_{c,t} + o_{tr,t} + o_{m,t} - y_{o,t}$.

The total trade balance is given by:

$$nx_t = nx_{f,t} - \frac{p_{o,t}}{p_{f,t}} (o_{c,t} + o_{tr,t} + o_{m,t} - y_{o,t}) \quad (40)$$

where nx_t is U.S. net exports to ROW. This depends on the value of net exports of final goods less the value of oil imports. If goods and oil markets are in equilibrium, the trade balance is equal to the net change in U.S. assets from the rest-of-the world ($nx_t = A_{t+1} - (1 + R_t)A_t$). Here we will assume that nx_t are zero in the steady state.

² In terms of percent deviations from the steady state: $\hat{y}_{GDP,t} = \frac{p_{f,ss} \hat{y}_{f,ss}}{ny_{GDP,ss}} + \frac{p_{o,ss} \hat{y}_{o,ss}}{ny_{GDP,ss}} - \frac{p_{o,ss} \hat{o}_{tr,ss}}{ny_{GDP,ss}} - \frac{p_{o,ss} \hat{o}_{m,ss}}{ny_{GDP,ss}}$ and $\hat{\pi}_{GDP,t} = \frac{p_{f,ss} \hat{\pi}_{f,ss}}{ny_{GDP,ss}} + \frac{p_{o,ss} (\hat{y}_{o,ss} - \hat{o}_{tr,ss} - \hat{o}_{m,ss})}{ny_{GDP,ss}} \hat{\pi}_{o,t}$. Nominal GDP in the steady state is given by: $ny_{GDP,ss} = p_{f,ss} y_{f,ss} + p_{o,ss} (y_{o,ss} - o_{tr,ss} - o_{m,ss})$.

2.8. Rest of the world

Rather than model the behavior of the ROW structurally, we model the ROW as reduced form equations. ROW oil supply is given by:

$$row_{s,t} = (row_{s,t-1})^{\hat{\theta}_s} \left(\left(\frac{p_{o,t}}{p_{f,t}^*} \right)^{\eta_s} z_{row,t} \right)^{1-\hat{\theta}_s} \quad (41)$$

where η_s is the long run price elasticity of supply, $\eta_s(1-\hat{\theta}_s)$ is the short run elasticity of supply, and $z_{row,t}$ is a supply shock. $\left(\frac{p_{o,t}}{p_{f,t}^*} \right)^{\eta_s}$ is the real price of oil in terms of the foreign final good. The parameter $\hat{\theta}_s$ reflects the inertia in the response of output to both price and supply shocks. ROW oil demand is given by:

$$row_{d,t} = (row_{d,t-1})^{\hat{\theta}_d} \left(\left(\frac{p_{o,t}}{p_{f,t}^*} \right)^{\eta_d} y_{row,t} z_{row,t}^d \right)^{1-\hat{\theta}_d} \quad (42)$$

where η_d is the long-run price elasticity of supply, $\eta_d(1-\hat{\theta}_d)$ is the short-run elasticity of supply, $y_{row,t}$ is ROW economic activity, η_y is the long-run income elasticity of oil demand, and $z_{row,t}^d$ is an exogenous oil specific demand shock.

For the evolution of ROW economic activity ($y_{row,t}$), U.S. net exports (nx_t), and the real exchange rate for the ROW ($\frac{p_{f,t}}{p_{f,t}^*}$), we specify a simple reduced form VARX(1) where the exogenous (“X”) variables are current and lagged values of real oil prices and U.S. production of final goods ($y_{f,t}$). The additional “X” variables represent the feedback to the rest-of-the-world from the world oil market and from the domestic U.S. economy. We assume a recursive causal structure among $y_{row,t}$, nx_t , and $\left(\frac{p_{f,t}}{p_{f,t}^*} \right)$ in that order. Thus, the ROW economic activity is given by:

$$\begin{aligned} \log \left(\frac{y_{row,t}}{y_{row,ss}} \right) &= a_{row}(L) \log \left(\frac{y_{row,t-1}}{y_{row,ss}} \right) + b_{row}(L) \log \left(\frac{nx_{t-1}}{nx_{ss}} \right) \\ &+ c_{row}(L) \log \left(\frac{p_{f,t-1}^*}{p_{f,t-1}} \right) + d_{row}(L) \log \left(\frac{p_{o,t}}{p_{f,t}} \right) \\ &+ e_{row}(L) \log \left(\frac{y_{f,t}}{y_{f,ss}} \right) + \varepsilon_{row,t} \end{aligned} \quad (43)$$

The term $\varepsilon_{row,t}$ represents an exogenous shock to world economic activity. U.S. net exports is given by:

$$\begin{aligned} \log \left(\frac{nx_t}{nx_{ss}} \right) &= a_{nx}(L) \log \left(\frac{y_{row,t}}{y_{row,ss}} \right) + b_{nx}(L) \log \left(\frac{nx_{t-1}}{nx_{ss}} \right) + c_{nx}(L) \log \left(\frac{p_{f,t-1}^*}{p_{f,t-1}} \right) \\ &+ d_{nx}(L) \log \left(\frac{p_{o,t}}{p_{f,t}} \right) + e_{nx}(L) \log \left(\frac{y_{f,t}}{y_{f,ss}} \right) + \varepsilon_{nx,t} \end{aligned} \quad (44)$$

The real exchange rate for the ROW:

$$\begin{aligned} \log \left(\frac{p_{f,t}^*}{p_{f,t}} \right) &= a_{rer}(L) \log \left(\frac{y_{row,t}}{y_{row,ss}} \right) + b_{rer}(L) \log \left(\frac{nx_t}{nx_{ss}} \right) + c_{rer}(L) \log \left(\frac{p_{f,t-1}^*}{p_{f,t-1}} \right) \\ &+ d_{rer}(L) \log \left(\frac{p_{o,t}}{p_{f,t}} \right) + e_{rer}(L) \log \left(\frac{y_{f,t}}{y_{f,ss}} \right) + \varepsilon_{rer,t} \end{aligned} \quad (45)$$

We assume the real exchange rate is one in the steady state.

2.9. Optimal oil usage

Because oil is used in similar ways across the consumption, transportation, and, intermediate goods sectors of the economy, we present for a general sector the conditions that describe the optimal use of oil and the efficiency choice. Because oil usage and efficiency are tied to capital services and capital in the model, we also present the optimality conditions for investment and the stock of capital.

The first order condition for oil usage is given by:

$$p_{j,t} \frac{\partial y_{j,t}}{\partial ks_{j,t}} \frac{\partial ks_{j,t}}{\partial (e_{j,t-1} o_{j,t})} e_{j,t-1} - p_{o,t} = 0 \quad (46)$$

for $j = c, tr, m$. This equates the value of the marginal product of oil with the price of oil where $p_{j,t}$ is the output price of good j , $\frac{\partial y_{j,t}}{\partial ks_{j,t}}$ is the marginal product of capital services while $\frac{\partial ks_{j,t}}{\partial (e_{j,t-1} o_{j,t})} e_{j,t-1}$ is the marginal product of oil usage in capital services. The optimal level of efficiency ($e_{j,t}$) is given by

$$M_{t,t+1} \left[p_{j,t+1} \frac{\partial y_{j,t+1}}{\partial ks_{j,t+1}} \frac{\partial ks_{j,t+1}}{\partial (e_{j,t} o_{j,t+1})} o_{j,t+1} + (1-\delta) \frac{e_{j,t+1} \lambda_{j,t+1}^e}{e_{j,t}} \right] - \lambda_{j,t}^e = 0 \quad (47)$$

where $\lambda_{e,t}$ is the shadow price of efficiency. Note we can combine the FOCs for oil and efficiency to obtain

$$M_{t,t+1} [p_{o,t+1} o_{j,t+1} + (1-\delta) e_{j,t+1} \lambda_{j,t+1}^e] - e_{j,t} \lambda_{j,t}^e = 0 \quad (48)$$

If one recursively substitutes, one can write the total value of efficiency as:

$$e_{j,t} \lambda_{j,t}^e = \sum_{i=1}^{\infty} (1-\delta)^i M_{t,t+i} p_{o,t+i} o_{j,t+i} + \lim_{i \rightarrow \infty} (1-\delta)^i M_{t,t+i} e_{j,t+i} \lambda_{j,t+i}^e \quad (49)$$

This says that the total value of efficiency is the discounted expenditures on oil.

Investment in efficiency, $x_{j,t}$, trades off the value derived from increasing efficiency and higher capital investment costs:

$$-p_{f,t} (1+\phi_I(I_{j,t}/k_{j,t-1})) \phi_e'(x_{j,t}) I_{j,t} + \delta \lambda_{j,t}^e \frac{e_{j,t}}{x_{j,t}} = 0 \quad (50)$$

where $p_{f,t} (1+\phi_I(I_{j,t}/k_{j,t-1})) \phi_e'(x_{j,t}) I_{j,t}$ is the increased capital investment costs of trying to improve efficiency. Note that in the steady state:

$$\lambda_{j,t}^e = \frac{\beta p_o o_j}{1-\beta(1-\delta)} \quad (51)$$

The shadow price of efficiency in the steady state is:

$$\lambda_j^e = p_f \phi_e'(x_{j,ss}) k_{j,ss} \quad (52)$$

Combining the first order of conditions of capital (not shown) with those for efficiency implies that in the steady state the value of efficiency relative to physical capital:

$$\frac{\lambda_j^e e_j}{\lambda_j^k k_j} = \frac{s_{o,j}}{1-s_{o,j}} \quad (53)$$

where $s_{o,j}$ is the expenditure share of oil in the production of capital services.

Making efficiency a choice variable dramatically changes the long-run elasticity of substitution of oil usage with respect to price changes. To see this, consider the short-run and long-run Allen Elasticities with respect to a permanent oil price change. Here we focus on capital services which we take to be a CES production function given by:

$$ks_{j,t} = z_{ksj,t} (\psi_{o,j} (e_{j,t-1} o_{j,t})^{\rho_{ksj}} + (1-\psi_{o,j}) k_{j,t-1}^{\rho_{ksj}})^{1/\rho_{ksj}} \quad (54)$$

The short-run Allen elasticity of oil (keeping total capital services fixed) is essentially zero:

$$\frac{do_j}{dp_o |_{dk_s=0, dk=0, de=0}} = 0 \quad (55)$$

Because capital and efficiency are predetermined, there is no scope for substitution in the short-run. The long-run (across steady states) Allen elasticity of oil for the standard case when efficiency is constant is:

$$\frac{do_j}{dp_o |_{dk_s=0, dk=0}} = -\frac{(1-s_{o,j})}{(1-\rho_{ksj})} \quad (56)$$

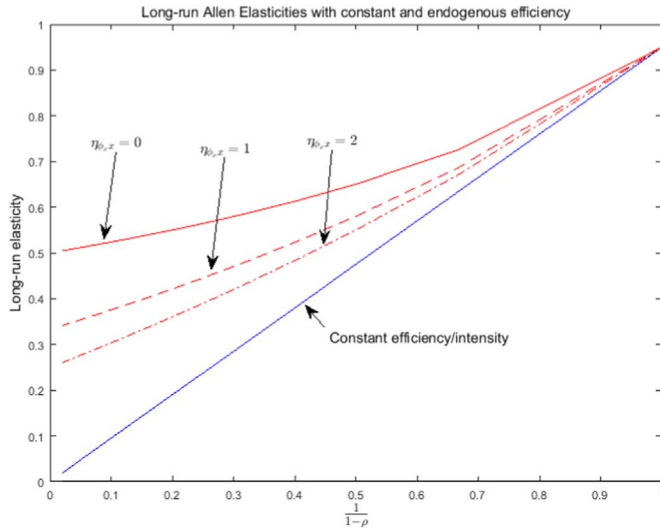


Fig. 1. Long-run Allen Elasticities with constant and endogenous efficiency.

When efficiency can be chosen along with oil and capital, the long-run Allen Elasticity for oil (keeping capital services and price of final goods constant) is:

$$\frac{d\sigma_j}{d\rho_{0|dk=0}} = - \frac{(1-s_{0,j}) - \frac{\rho_{ksj}}{1+\eta_{px}}}{(1-\rho_{ksj}) - \frac{\rho_{ksj}}{1+\eta_{px}}} \quad (57)$$

where $\eta_{px} = \frac{\varphi''(x_{j,ss})x_{j,ss}}{\varphi'(x_{j,ss})}$ is elasticity of the marginal cost of increasing efficiency. Note that as long as $\rho_{ksj} < 0$, the long-run response of oil usage to price change is larger when efficiency can change over time. The difference between the case when efficiency is constant and when it is chosen becomes greater the less oil and capital are substitutes (with ρ_{ksj} becoming more negative); while this difference gets smaller, the larger (in absolute value) is η_{px} . Fig. 1 displays how the long-run price elasticity of oil demand changes with respect to the elasticity of the marginal costs of changing oil intensity. Lower costs increase the long-run elasticity compared to the case where oil intensity is constant.

3. Estimation

Taking the first order conditions of household utility maximization and firms' profit maximization, and the market clearing conditions, we (log) linearize the model around the deterministic steady state.³ This yields a linear rational expectations model whose solution can be written in linear state-space form:

$$Y_t = H(\theta)S_t \quad (58)$$

$$S_t = F(\theta)S_{t-1} + v_t \quad (59)$$

where Y_t is a vector of endogenous control variables, S_t is a vector of (possibly unobserved) state variables, and v_t is a vector of exogenous structural shocks with $v_t \sim N(0, Q(\theta))$. The elements in the matrices $H(\theta)$, $F(\theta)$, and $Q(\theta)$ depend on the structural parameters in the model, denoted by θ . We use only a subset of the variables in our estimation. These are given by vector, Y_t^{obs} , so the observation equation in estimation is given by:

$$Y_t^{obs} = H^{obs}(\theta)S_t \quad (60)$$

Given the data, the predictive likelihood for a given parameter vector and the data, Y_T , is denoted by $L(\theta, Y_T)$ and can be obtained by the Kalman filter.

³ We use a combination of Dynare subroutines and custom written Matlab code to solve and estimate the model.

Given a prior distribution over the parameters, $p(\theta)$, the posterior distribution of the parameters satisfies

$$P(\theta|Y_T) \propto L(\theta, Y_T)p(\theta) \quad (61)$$

We estimate the posterior distribution of the parameters by sampling from a random walk Metropolis-Hastings Monte Carlo Markov Chain (MH-MCMC).⁴ We “tune” the Metropolis-Hastings draws over the first 500,000 draws and then use the last 500,000 draws as our sample of realizations from the posterior distribution of the parameters. This sample of draws from the posterior distribution can be used to estimate the posterior distribution of any function of the underlying parameters such as impulse responses and, in particular, the elasticity of real GDP with respect to oil price changes.

3.1. Shocks and observables

In our benchmark model, we include 13 variables in our observation vector and 13 structural shocks. Our choice of shocks and observables was guided by our desire to estimate the output elasticity of U.S. real GDP with respect to real oil price changes brought about by oil supply shocks in the ROW. Furthermore, as we are interested in assessing oil usage in the transportation sector, we include also observables and shocks pertaining to transportation. Table 1 lists the variables included in the observation vector. The U.S. macro variables include: log real GDP per capita, log total hours per capita, inflation as measured by the GDP deflator, and the ex-post real interest rate (three-month Treasury bill minus inflation). Oil related U.S. variables include: log of the share of gasoline consumption in nominal GDP, log of the share of investment in transportation equipment in nominal GDP, log of petroleum imports as a share of GDP, and the log of U.S. oil production. World oil market variables include: log real price of oil (deflated by the U.S. GDP deflator) and the log of ROW oil production. International trade variables include: world industrial production, real exchange rate for foreign goods (relative price of foreign final goods in terms of U.S. final goods), and U.S. net exports as a percent of nominal GDP.⁵

We use quarterly data from first quarter 1991 through fourth quarter 2015. We were reluctant to start our sample earlier given the structural changes that have occurred in the oil market and U.S. economy.⁶ We judged this sample to be a good tradeoff between having sufficient sample size to estimate the parameters and having relatively few structural changes. As our focus was on more of the short-run consequences of oil price shocks and model solution was based on deviations from a steady state (or balanced growth path), variables with a substantial secular trend, U.S. real GDP, U.S. hours, U.S. oil production, ROW oil production, and ROW economic activity, were detrended with deterministic, quadratic time trends; the other variables were demeaned.

To control for other sources of economic fluctuations other than oil supply shocks, we include a variety of U.S. macro shocks (representing different shocks to U.S. oil demand), ROW shocks to oil supply and oil specific demand shocks, ROW economic activity and trade shocks, and, finally, sectoral shocks (TFP shocks to U.S. intermediate goods and U.S. oil production). The specific structural shocks we include are: TFP shock in the intermediate goods sector ($z_{m,t}$), shock to total consumption (or, equivalently, preference shock), monetary policy shock, risk shock, aggregate investment productivity shock, U.S. oil production

⁴ We restrict the parameter space so that the Blanchard and Kahn (1980) (BK) condition for a unique, stable solution holds. In practice, this is done by rejecting parameter draws in which the BK condition fails to hold. This means that the posterior density is quite complicated with multiple modes, cliffs, and ridges. This in turn makes exploration of the parameter space quite challenging.

⁵ Most of the data were obtained from The Federal Reserve Bank of St. Louis' Fed® database. The world industrial production series is from CPB Netherlands Bureau for Economic Policy Analysis and excludes construction activity.

⁶ According to Huntington (2018), the estimation period includes four major ROW oil supply disruptions.

Table 1
List of observables and structural shocks.

Observable variable		Structural shock	
1.	log(per capita US real GDP) (detrended)	1.	Productivity shock to goods, $z_{m,t}$
2.	log(per capita hours worked) (detrended)	2.	Productivity/preference shock to consumption, $z_{c,t}$
3.	Ex-post real interest rate	3.	US monetary policy shock
4.	log(US personal consumption expenditures on gasoline as a % of GDP)	4.	US risk shock: $R_t = R_t^{policy} z_{risk,t}$
5.	log(US transportation equipment investment expenditures as a % of GDP)	5.	Productivity shock to investment, $z_{I,t}$
6.	log(US oil production) (detrended)	6.	Productivity shock to domestic oil production, $z_{o,t}$
7.	log(US petroleum imports as % of GDP)	7.	US price mark-up shock
8.	log(real price of oil)	8.	ROW oil supply shock
9.	log(ROW oil production) (detrended)	9.	ROW oil specific demand shock
10.	log(world economic activity) (detrended)	10.	ROW shock to economic activity
11.	log(real exchange rate-trade weighted)	11.	ROW real exchange rate shock
12.	US net exports as a % of GDP	12.	ROW shock to US net exports
13.	Inflation rate (GDP deflator)	13.	Inflation target in Taylor rule

productivity shock, U.S. price markup shock, a shock to the inflation target in the U.S. monetary policy rule, ROW oil supply shock, ROW oil specific demand shock, and ROW reduced form shocks to world economic activity, ROW real exchange rate, and U.S. net exports. While including more observables and shocks would have been desirable, we believe that the technical burden of estimating a larger model would have more than offset the benefit of increasing the size of the estimated model. We judged this set of observables and shocks to be sufficient to get the interaction between the U.S. economy and oil but allow for other international elements outside of the oil market to influence U.S. economic activity and the oil market.

3.2. Prior distribution of parameters

We can break the parameters in our model into roughly three groups: (1) calibrated parameters; (2) parameters over which we have informed priors; and (3) parameters over which we have relatively uninformed priors. Table 2 lists the calibrated parameters. The key share parameters that we set equal to their sample averages are: share of gasoline expenditures in consumption (this is a proxy for the share of oil consumption in total consumption), share of automobile purchases in total consumption (proxy for share of capital in capital/transportation services used in consumption), the share of the transportation sector in aggregate gross output (we abstract from other inputs and approximate gross output as valued added plus transportation services).

Table 2
Calibrated share parameters.

Parameters	value
discount factor in utility (β)	0.99
depreciation rate (δ)	0.025
US Consumption and labor shares:	
share of gasoline in total consumption	0.03
share of motor vehicle expenditures in total consumption	0.045
share of labor in total hours	0.30
US Production shares:	
share of transport in aggregate gross output	0.03
share of labor in gross output of transport sector	0.50
share of energy input in transport gross output	0.25
share of labor in intermediate goods	0.60
share of petroleum products as inputs in intermediate goods	0.015
share of labor in value added of oil and gas extraction	0.15
US vs ROW shares	
share of oil imports in total US oil consumption	0.67
share of US oil production in total world production	0.08
Implied shares:	
share of transportation in total US oil consumption	0.67
share of oil expenditures in GDP	.035

Similarly, we calculate labor share in gross output in transportation and oil extraction sectors. We also use the share of U.S. oil imports in total U.S. oil consumption and the share of U.S. oil production in total world oil production to calibrate the relative sizes of the U.S. and ROW oil supplies and demands. The ratio of U.S. oil consumption expenditures to nominal GDP and the ratio oil consumption in transportation services to total U.S. oil consumption ($\frac{o_{c,t} + o_{tr,t}}{o_{c,t} + o_{tr,t} + o_{m,t}}$) are used to set the shares of oil consumption in goods production and transportation.

The second group of parameters are estimated, and we have

Table 3
Prior distribution of key parameters.

Parameter	Prior distribution	5th	Mode	95th
Macro parameters				
Inverse Frisch elasticity (η)	Gamma(2,0.5)	0.18	0.5	2.37
Inverse intermp. Elast. (σ)	Gamma(4,0.5)	0.98	2	4.58
Habits weight (χ_c)	Beta(0.65,1.5,0,1)	0.19	0.7	0.93
Habits persistence (γ_h)	Beta(0.975,1.75,0,1)	0.04	0.1	0.83
Inflation weight in Taylor rule (γ_π)	Beta(3,3,1.0,2.0)	1.29	1.5	1.71
Output weight in Taylor rule (γ_y)	Beta(5,3,0,1)	0.17	0.3	0.49
Taylor rule inertia (γ_R)	Beta(1.05,1.05,0,1)	0.06	0.5	0.94
Real wage persistence (γ_w)	Beta(0.65,1.5,0,1)	0.19	0.7	0.93
Rotemberg price adjustment (α_p)	Gamma(3,50)	40.9	100	314.8
Rotemberg price indexation	Beta(0.975,1.75,0,0.5)	0.02	0.05	0.41
Shock processes				
Autoregressive parameters	Beta(1.05,1.05,0,1)	0.06	0.5	0.94
Standard deviation	Gamma(0.067,1.3)	0.05	0.1	1.08
Sectoral parameters				
Elasticity of substitution ($\frac{1}{1-\rho_j}$), $j = c, f, tr, o$	Gamma(0.67,1.3)	0.47	1.0	2.47
Elasticity of substitution ($\frac{1}{1-\rho_j}$), $j = ksc, kstr, ksm$	Gamma(0.067,1.3)	0.05	0.1	1.08
Labor, capital, oil intensity adjustment costs	Gamma(1.5100)	17.6	50	390.7
ROW parameters				
(-1)xROW long-run oil demand elasticity	Beta(2.2,4,0.4,1.5)	0.49	0.7	1.14
ROW long-run oil supply elasticity	Beta(2.2,4,0.4,1.5)	0.49	0.7	1.14
(-1)xROW short-run oil demand elasticity	Beta(1.2,4,0.4,1.5)	0.03	0.2	0.37
ROW short-run oil supply elasticity	Beta(1.2,4,0.4,1.5)	0.03	0.2	0.37
ROW oil demand income elasticity	Normal(0.7,0.25)	-0.12	0.7	1.52

Table 4
Posterior distributions of key parameters.

Parameter	Mode	Median	Mean	5th	95th
Panel A					
Macro parameters:					
Inverse Frisch elasticity (η)	1.49	1.32	1.30	0.58	2.13
Inverse intertemporal elast of substitution (σ)	1.72	1.66	1.73	0.87	2.91
Habits weight (χ_c)	0.99	0.99	0.99	0.98	1.00
Habits persistence (γ_h)	0.31	0.20	0.21	0.10	0.37
Inflation weight in Taylor rule (γ_π)	1.56	1.30	1.30	1.16	1.49
Output weight in Taylor rule (γ_y)	0.42	0.35	0.35	0.21	0.46
Taylor rule inertia (γ_R)	0.95	0.95	0.94	0.92	0.96
Real wage persistence (γ_w)	0.04	0.04	0.05	0.03	0.09
Rotemberg price adjustment (α_p)	222.9	207.1	205.7	140.8	255.4
Rotemberg price indexation	0.002	0.006	0.007	0.001	0.015
Sectoral parameters:					
Elasticity of substitution: nonoil and oil related durable services in consumption	4.75	5.57	5.59	4.26	6.95
Elasticity of substitution: transportation and goods in final goods	0.41	0.38	0.38	0.29	0.48
Elasticity of substitution: labor and capital services in transportation	0.53	0.67	0.69	0.41	1.03
Elasticity of substitution: labor and capital in oil production	1.44	1.27	1.34	0.76	2.15
Elasticity of substitution: oil and capital in consumption durables services	0.45	0.45	0.45	0.42	0.49
Elasticity of substitution: oil and capital in transportation capital services	0.32	0.34	0.34	0.24	0.46
Elasticity of substitution: oil and capital in goods capital services	1.44	1.21	1.22	0.98	1.52
ROW parameters					
(-1)×ROW long-run oil demand elasticity	0.43	0.49	0.51	0.42	0.65
ROW long-run oil supply elasticity	0.52	0.53	0.55	0.43	0.75
(-1)×ROW short-run oil demand elasticity	0.15	0.18	0.18	0.14	0.23
ROW short-run oil supply elasticity	0.003	0.005	0.005	0.003	0.007
ROW long-run oil demand income elasticity	0.96	0.98	1.02	0.59	1.62
Panel B					
Labor: goods	0.004	0.01	0.02	0.003	0.06
Labor: transportation	36.7	33.1	33.3	16.6	51.4
Labor: oil	8.63	9.69	12.6	4.75	36.3
Capital: oil intensive consumer durables	142.6	143.1	145.8	122.5	176.3
Capital: goods	3130.6	3012.1	3022.1	2427.0	3632.6
Capital: transportation	977.6	998.0	1002.9	87.3	1147.5
Capital: oil	2789.4	2361.2	2402.9	1922.2	3022.4
Oil intensity/efficiency: oil intensive durables	130.8	82.3	86.6	25.0	175.6
Oil intensity/efficiency: transportation	161.4	263.2	280.1	80.6	506.2
Oil intensity/efficiency: goods	32.6	55.2	63.9	21.5	130.2
Autoregressive coefficient on exogenous shocks					
Intermediate goods TFP	0.98	0.96	0.96	0.92	0.98
Consumption	0.89	0.90	0.90	0.87	0.94
Investment technology	0.70	0.70	0.70	0.65	0.73
US oil technology	0.90	0.90	0.90	0.87	0.94
US markup	0.98	0.99	0.99	0.97	0.99
US risk	0.85	0.90	0.90	0.86	0.96
ROW oil supply	0.03	0.07	0.07	0.02	0.15
ROW oil demand	0.90	0.89	0.89	0.85	0.93
US. Inflation target	0.98	0.97	0.97	0.96	0.98
Standard deviations of exogenous shocks					
Intermediate goods TFP	0.005	0.005	0.005	0.004	0.006
Consumption	0.034	0.035	0.35	0.030	0.041

Table 4 (continued)

Parameter	Mode	Median	Mean	5th	95th
Investment technology	0.226	0.218	0.220	0.182	0.264
US oil technology	0.027	0.027	0.027	0.024	0.030
US markup	0.079	0.54	0.056	0.038	0.085
US risk	0.010	0.008	0.007	0.003	0.011
ROW oil supply	1.72	1.28	1.31	0.82	1.92
ROW oil demand	0.103	0.110	0.112	0.087	0.146
US interest rate rule	0.0014	0.0013	0.0013	0.0011	0.0016
ROW economic activity	0.009	0.009	0.009	0.008	0.010
Real exchange rate	0.018	0.018	0.018	0.016	0.020
Net exports	0.111	0.105	0.106	0.090	0.126
US. Inflation target	0.005	0.006	0.007	0.005	0.010

relatively informed priors (Table 3). For the macro parameters, we set prior modes to values common in the literature. For the ROW oil supply and demand elasticities, we set generalized beta distributions with a mode of 0.7 for the long-run and 0.15 for the short-run elasticities of supply and -0.7 for the long-run and -0.15 for the short-run elasticities of demand. The parameters governing the stochastic processes of the exogenous shocks as well as the ROW trade variables are relatively uninformed.

4. Empirical results

4.1. Parameter estimates

Table 4, Panels A and B display the posterior distribution of some of the key parameters for the baseline model. Most of the “macro” parameters are well within the range implied by typical macro models. The exceptions include the real wage rigidity which was estimated on the low side of the prior distribution and the indexation parameters in the Rotemberg price adjustment which is essentially estimated to be zero.

Sectoral elasticities of substitution between nondurable and oil-related consumption services were estimated to be well above one while the elasticity of substitution between labor and capital in oil production and the elasticity of substitution between oil and capital in production of goods capital services were slightly above one. The remaining elasticities of substitution were estimated to be below one. Weighted by the steady state shares in U.S. oil consumption, these oil/capital elasticities of substitution yield an aggregate oil/capital elasticity of substitution of around 0.65.

The ROW oil supply and demand elasticities are presented at the bottom of Table 4, Panel A. The long-run price elasticity of oil supply and demand are roughly equal, with posterior median/mean of around 0.5 and -0.5, respectively. The short-run price elasticity of ROW oil demand is substantially larger than that of ROW oil supply suggesting very slow adjustment of ROW oil production to oil price changes. This difference has implications for attributing the sources of short-run fluctuations in oil prices as we shall see in the next section.

Table 4, Panel B reports the structural parameters associated with adjustment costs as well as parameters of the stochastic processes for the exogenous shocks. Most of the adjustment cost parameters suggest substantial slow adjustment in labor, capital, and oil intensity; the exceptions being labor and efficiency in the transportation sector and oil using consumer durables in consumption.

Fig. 2 displays the actual and one step ahead predicted values implied by the model evaluated at the posterior mode of the parameters. Recall that the actual data have been logged and either detrended or demeaned (see Table 1). The predicted value is from the Kalman filter and forms the basis for the likelihood function of the model ($Y_{t|t-1}^{obs} = H^{obs}(\theta)S_{t|t-1}$) where θ is set equal to the posterior mode. From Fig. 2, one notes the predicted values from the model track the observed variables pretty well (albeit typically with a lag). The model appears to

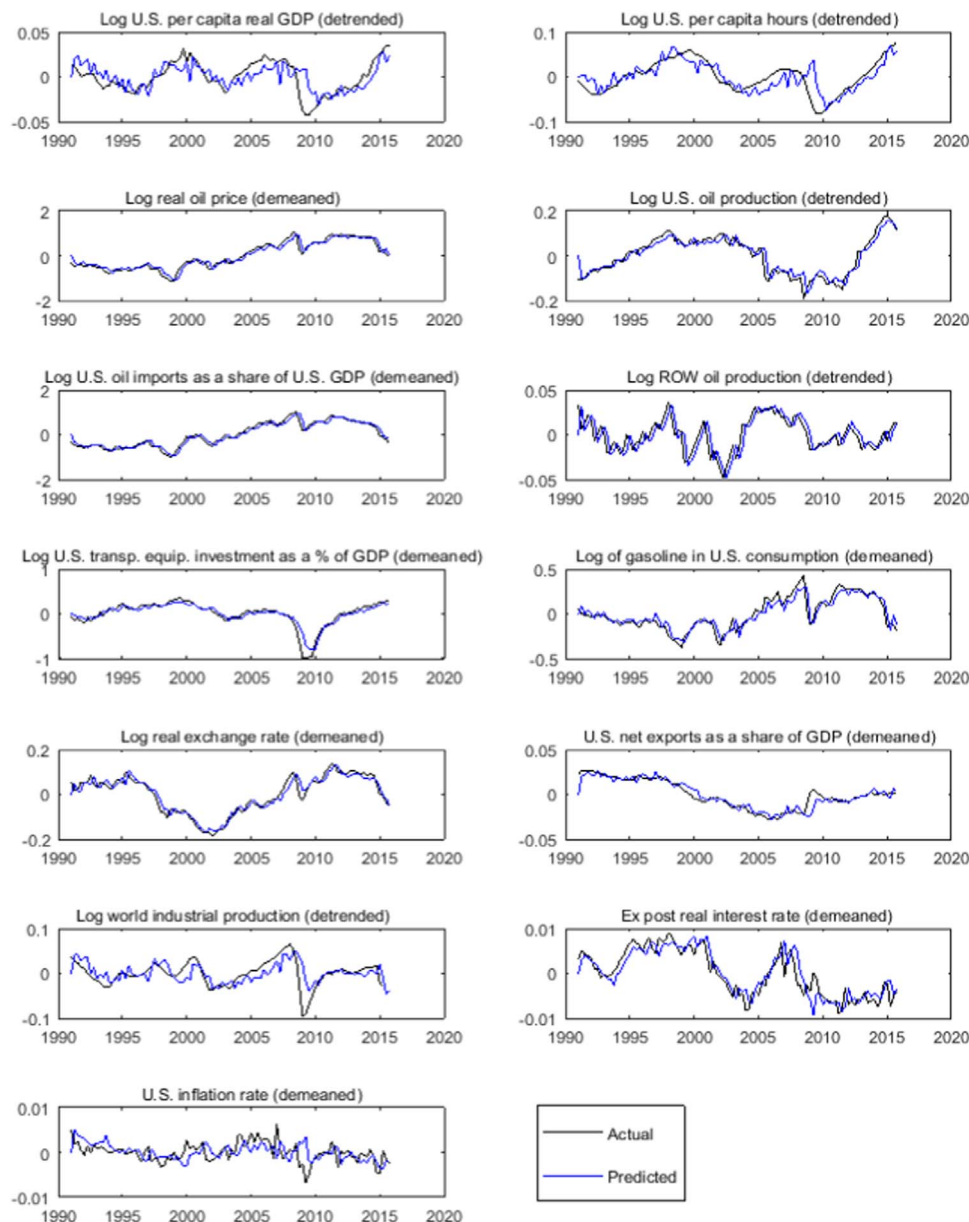


Fig. 2. Actual and one step ahead predicted value from model.

face a trade-off between “fitting” inflation and interest rates versus U.S. real GDP and hours in periods around the Great Recession.⁷

4.2. Impulse response analysis

While space constraints prevent us from presenting the Impulse Response Functions (IRFs) for all thirteen of the shocks, we do present the impulse responses of selected variables to shocks originating in the oil market. Fig. 3 displays the impulse responses of selected variables to a one standard deviation negative shock to ROW oil supply. We find that following a ROW supply shock, the real oil price rises and then slowly return to its pre-shock level. U.S. real GDP falls on impact and then recovers in response to the shock. U.S. hours also fall in response to ROW oil supply shock and subsequently recover. Not surprisingly,

ROW oil output falls in response to supply shock, but given the inertia estimated in ROW supply the maximum decline in ROW oil output occurs about one quarter after the shock. On the other hand, U.S. oil supply rises in response to increases in oil prices brought about by the decline in ROW oil supply. This response peaks around seven quarters after the shock and is relatively small with an implied U.S. oil supply elasticity in the first year after the shock of around 0.62. Finally, the quantity of U.S. oil imports falls in response to the increase in oil prices with implied elasticity of close to negative one over the course of a year.

Fig. 4 shows the IRFs of select variables to a ROW oil-specific demand shock. As shown in the figure, the peak increase in oil prices occurs two-to-three quarters after the shock owing to inertia in ROW demand. U.S. hours worked and real GDP contract on impact, but the interior 90% of the posterior distribution includes the value of zero. Furthermore, the subsequent response is positive albeit quantitatively small (with an implied elasticity between -0.001 and 0.002 depending on the horizon). Overall, ROW oil specific demand shocks differ from ROW oil supply shocks in their effects on U.S. economic activity, a finding consistent with Baumeister and Hamilton (2015). Consistent

⁷ In a previous version of this paper, we included only the ex-post real interest rate and not inflation as observables. In that case, the predictive errors for real GDP and hours were slightly smaller than in the current version of the model but the predictive error for the real interest rate was substantially larger.

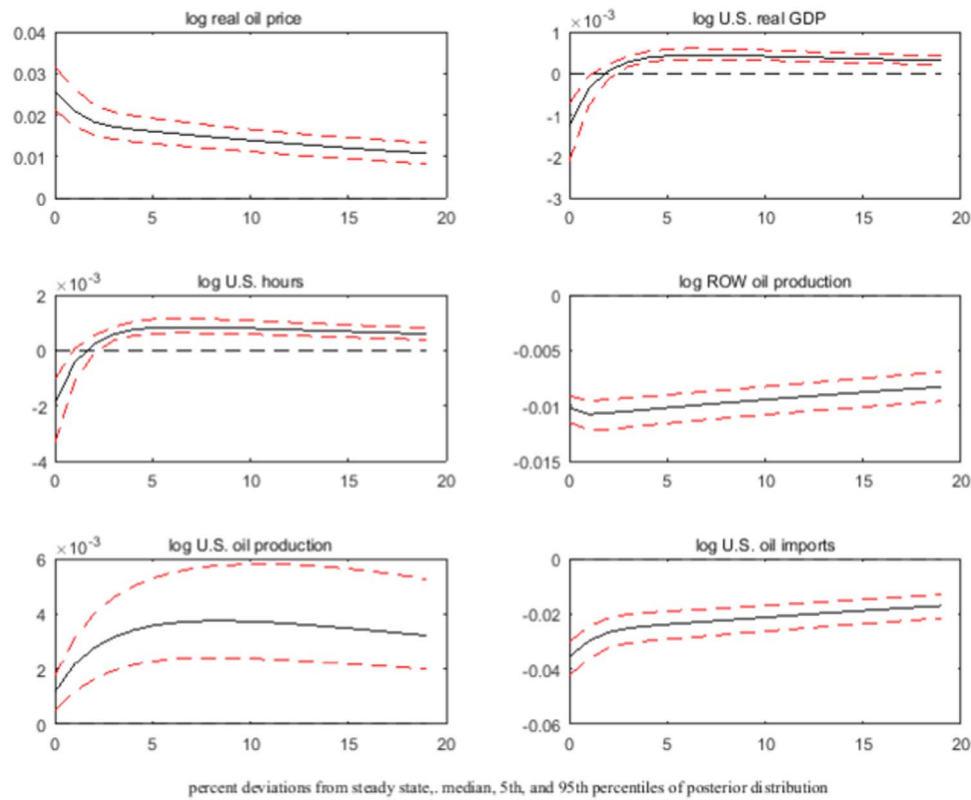


Fig. 3. Response to one standard deviation rest of world oil supply shock.

with a standard neoclassical effect, U.S. consumers appear to seek to maintain consumption—with both U.S. hours worked and real GDP increasing. ROW and U.S. oil production rises in response to the oil

specific demand shock, with U.S. oil supply increasing more in percentage terms than ROW supply.

Fig. 5 displays the responses to a positive technology shock to the

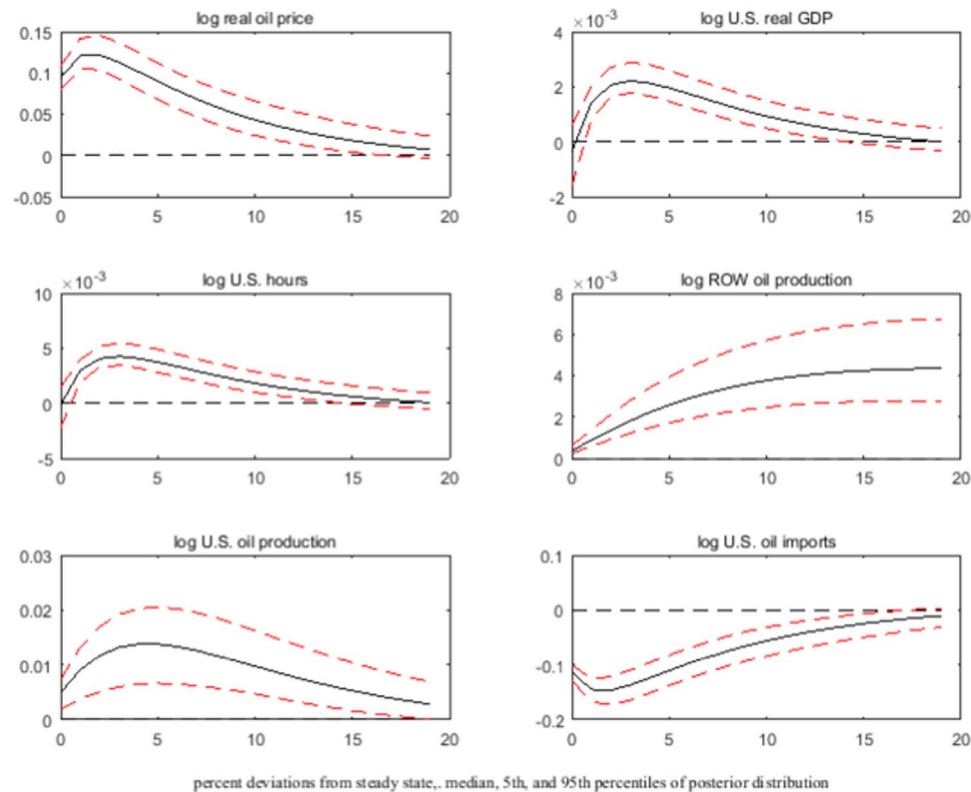


Fig. 4. Response to a one standard deviation rest of world oil specific demand shock.

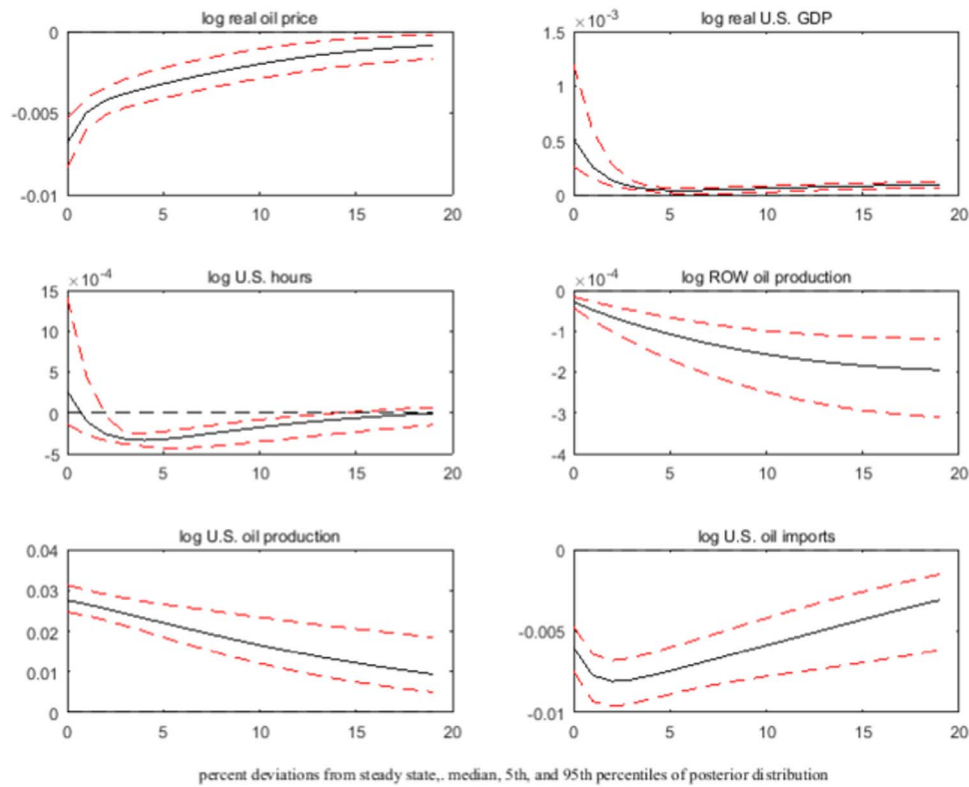


Fig. 5. Response to a one standard deviation U.S. oil production productivity shock.

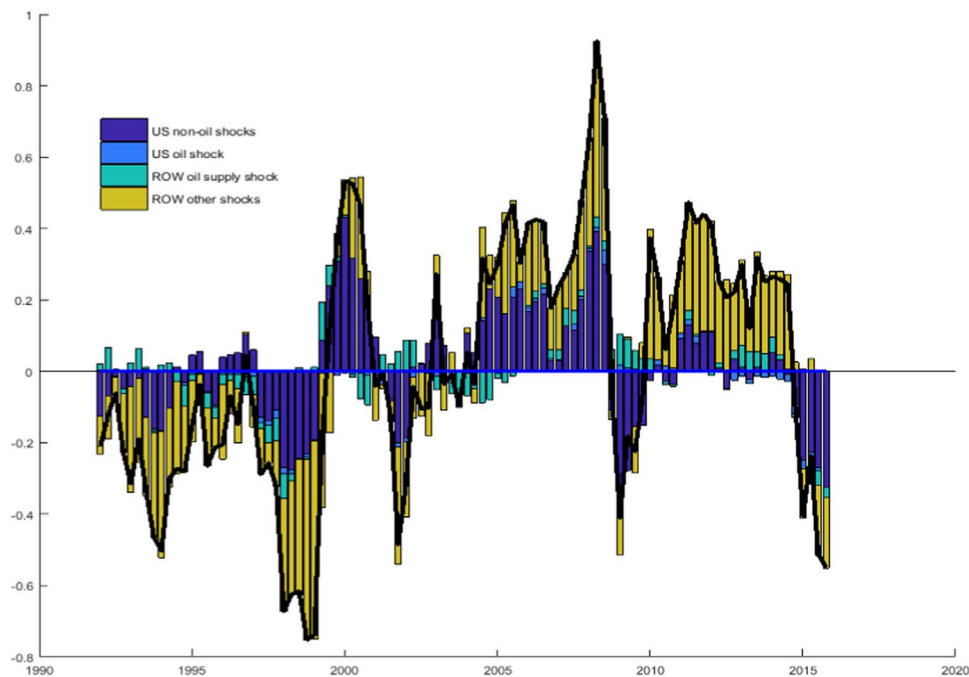


Fig. 6. Decomposition of one year ahead forecast error for real (log) oil price.

U.S. oil production sector. As expected, U.S. oil production increases as a result of this shock. U.S. hours worked and real GDP increase as well while world oil price, ROW oil production, and U.S. oil imports all fall.

4.3. Historical decompositions

Fig. 6 displays the historical decomposition of real oil price fluctuations. Specifically, the figure displays the contributions of various

groups of shocks to the forecast error implied by the model (based on the smoothed states) over a four-quarter horizon.⁸ Given the relatively large number of shocks, we combine these into four groups: non-oil

⁸ The implied forecasted error based on the smoothed states is $Y_{t+4}^{obs} - H^{obs}(\theta)F(\theta)^4 S_{t|T}$. $S_{t|T}$ is the estimated state vector at time t based on the full sample and is obtained from the Kalman smoother. This forecast error is decomposed into sources from the respective structural shocks.

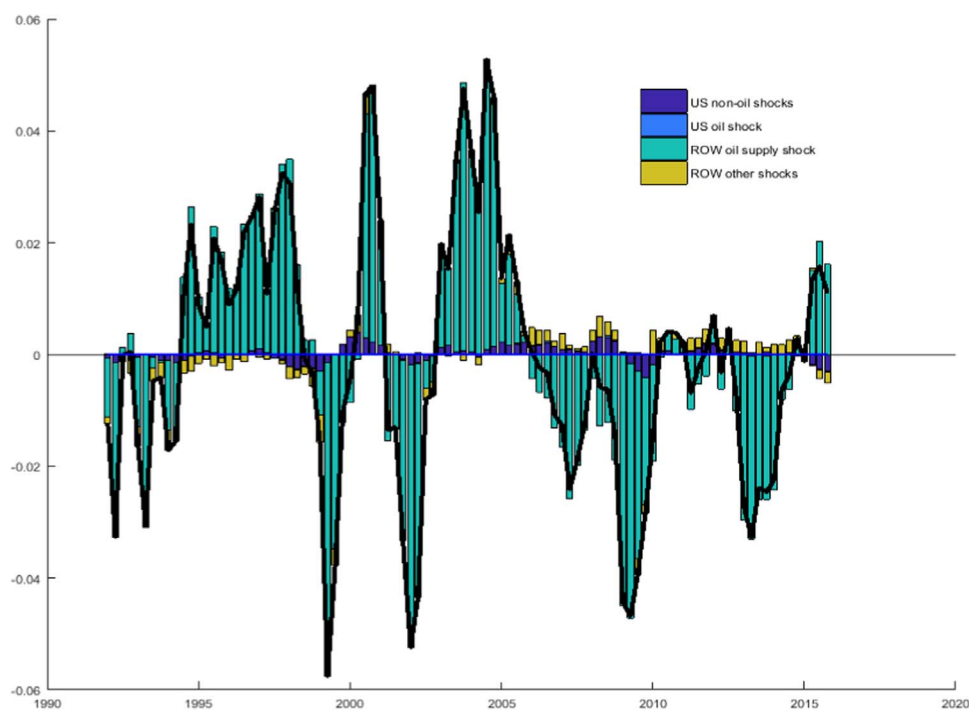


Fig. 7. Decomposition of one year ahead forecast of (log) rest of world oil production.

shocks originating in the United States (risk, interest rate rule, price markup, consumption, inflation target, goods TFP and investment shocks), oil supply productivity shock in the United States, ROW oil supply shocks, ROW shocks (oil specific demand, ROW economic activity, real exchange rate, and net exports shocks).

As shown in the figure, a large fraction of oil price fluctuations over a one-year horizon appears to be driven by ROW oil demand shocks, either oil specific demand shocks or shocks that affect ROW oil demand through ROW economic activity. The run up in oil prices in the mid-2000s, the collapse and recovery in oil prices after 2009–10, and the recent decline in 2014–15 were all attributed largely to ROW demand shocks. Over our sample, ROW oil supply shocks do contribute to oil price fluctuations but not nearly to the extent that ROW demand shocks do. During the 2006–2015 period, ROW oil supply shock's contribution to oil price movements was positive. According to the model, U.S. shocks also contribute modestly to oil price fluctuations. U.S. oil supply shocks contribute to a decline in oil prices in the 2011–2014 period, but their contribution is relatively modest.

Figs. 7 and 8 display the historical decompositions for ROW oil production and U.S. oil production, respectively. In contrast to oil prices, oil production fluctuations over the course of a year tend to be driven primarily by oil supply shocks. For ROW oil production, the greater than expected oil production in the 1990s and the lower than expected production starting in 2006 were driven by oil supply shocks. For U.S. oil production, the effects of the “Shale Revolution” show up as large surprises in U.S. oil production largely due U.S. oil production (technology) shocks.

The finding that oil demand shocks (originating in the ROW and the US) appear to drive most of the real oil price fluctuations, while oil supply shocks drive oil output fluctuations, is consistent with our estimates of ROW oil demand and supply price elasticities presented in Table 4.⁹ The short-run ROW oil demand price elasticity was estimated to be in the -0.18 range while the short-run ROW oil supply price elasticity was estimated to be in the 0.01 range. Thus, demand shocks, everything else equal, result in larger

price movements and smaller quantity movements than supply shocks at short horizons (here one year).¹⁰

Finally, Fig. 9 displays the historical decomposition for four quarter forecast error for U.S. real GDP. Fig. 9 suggests that annual fluctuations in U.S. real GDP over our sample were largely due to non-oil shocks originating in the United States. Although four major oil supply disruptions occurred during our estimation period, ROW oil supply shocks do not seem to have been a major contributor to U.S. real GDP fluctuations. These results are consistent with Christiano et al. (2014) who find that risk premium shocks were largely responsible for the decline in output during the 2007–2009 recession and that oil price fluctuations had relatively small effects on U.S. GDP fluctuations. We do find, however, that U.S. oil supply shocks provided a small positive contribution to U.S. real GDP during the 2010–2014 period.

5. GDP/Oil elasticities

Because real GDP/Oil price elasticities play a prominent role in macroeconomic/oil policy discussions, Table 5 displays GDP/Oil price elasticities for the baseline model and where we calibrate rather than estimate the macro parameters of the model. For the baseline model, the posterior mode, median, and mean values of the GDP/oil price elasticity are -0.022 , -0.015 , and -0.017 , respectively. The interior 90% posterior interval is $[-0.032 - 0.002]$. These values are among the least elastic in the older empirical macroeconomics oil price literature (as documented by Jones et al., 2004, Leiby, 2008, Brown and Huntington, 2013, 2015 and Brown, 2018). Nonetheless, recent estimates with DSGE models, such as Balke et al. (2010) and Bodenstein and Guerrieri (2011) and Kormilitsina (2011) find point estimates of the GDP/oil price elasticity of -0.006 , -0.012 , and -0.006 , respectively.¹¹

¹⁰ This sensitivity was also noted by Caldara, Cavallo, and Iacoviello (2016). In their structural VAR analysis, they find more equal supply and demand elasticities and, hence, a larger contribution of supply shocks to oil prices and a larger contribution of demand shocks to oil production.

¹¹ Kilian (2009), Herrera and Pesavento (2009), Blanchard and Gali (2010), Kilian and Vigfusson (2011a) (2011b), and Baumeister and Hamilton (2015) find similarly low values with structural vector autoregressive models.

⁹ Bodenstein and Guerrieri (2011) also find that oil price fluctuations were driven primarily by ROW oil efficiency shocks which would correspond to an oil specific demand shock in our framework.

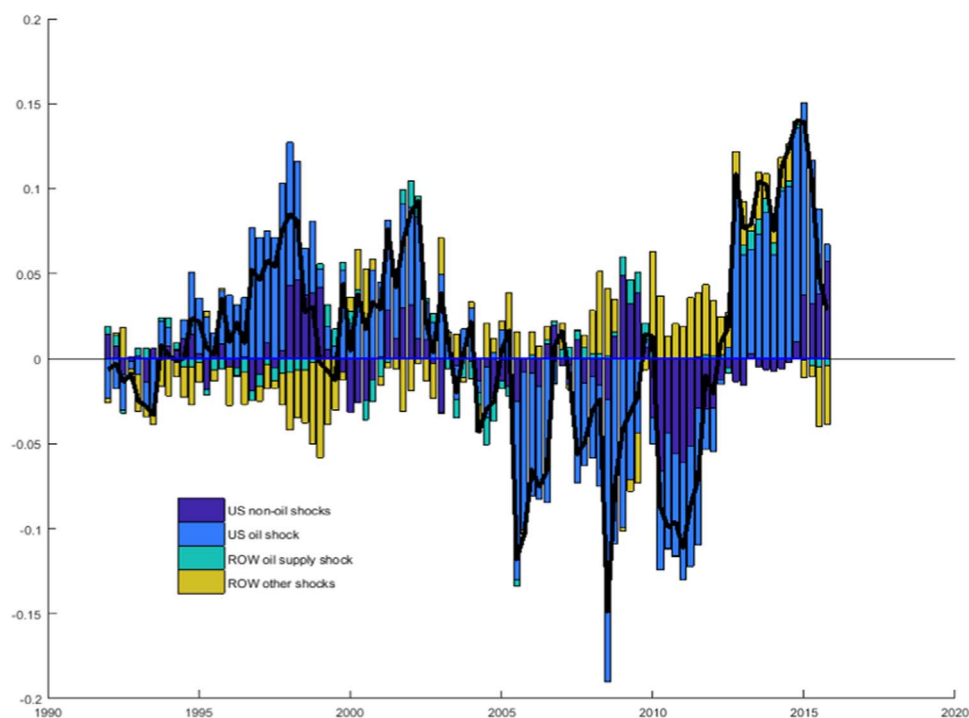


Fig. 8. Decomposition of one year ahead forecast error for (log) U.S. oil production.

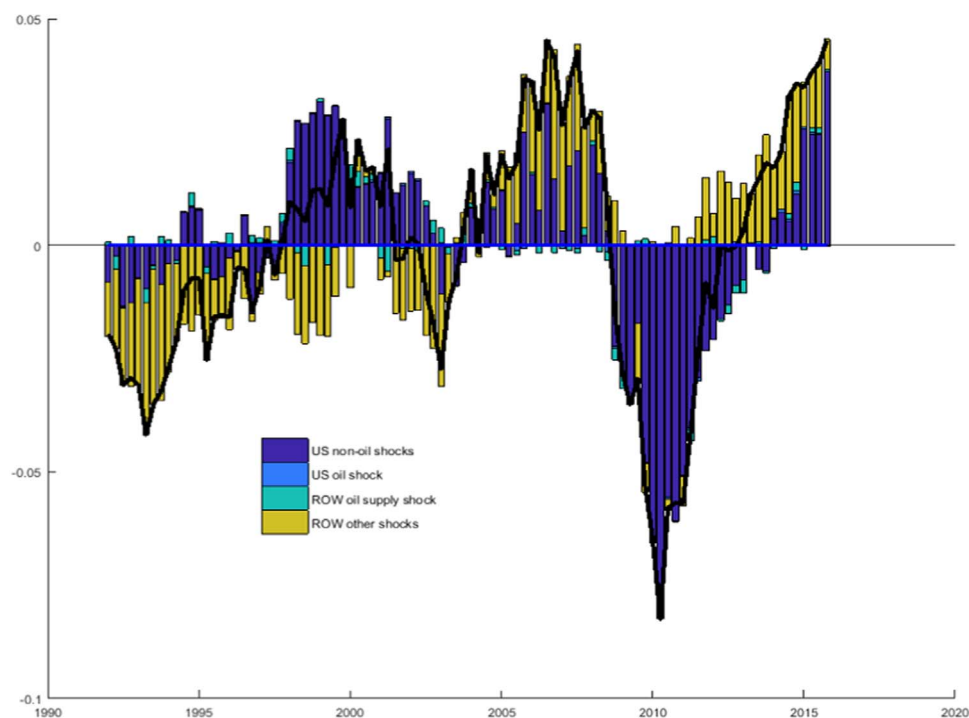


Fig. 9. Decomposition of one year ahead forecast error for (log) U.S. real GDP.

Table 5
Estimated GDP-Oil price elasticities due to rest of world oil supply shock.

Posterior distribution	Baseline model	Calibrated macro parameters
Mode	−0.022	−0.004
Median	−0.015	−0.010
Mean	−0.017	−0.010
5th percentile	−0.032	−0.015
95th percentile	−0.002	−0.003

To test the sensitivity of the model to alternative modeling or estimation choices, we also estimate a version of the baseline model where we set the macro parameters equal to the modes of their prior distributions. In this case, mode of the GDP/oil elasticity is estimated as less responsive than the baseline case (at -0.010).

What accounts for the relatively unresponsive GDP/oil price elasticities implied by the estimated DSGE models? To get a sense of the channels through which oil supply shocks affects economic activity in our model, consider a first order (log) linear approximation around the steady state for real GDP:

$$\hat{y}_t^{GDP} = \frac{1}{(1-s_o + s_y^o)} (\hat{y}_t^f - s_o \hat{o}_t^y + s_y^o \hat{y}_t^o) \quad (62)$$

where $\hat{y}_t^{GDP}$ is deviations of log real GDP from the steady state, $\hat{y}_t^f$ is (log) output of final good or gross output, $\hat{o}_t^y$ is the (log) oil used in production of the final good, $\hat{y}_t^o$ is the (log) domestic production of oil, s_o is the share of oil in the production of final good, and s_y^o is the ratio of the value of domestic oil production to the value of the final good. Substituting for final goods, Eq. (62) can be rewritten as:

$$\hat{y}_t^{GDP} = \frac{1}{(1-s_o + s_y^o)} (s_l \hat{l}_t^y + s_k \hat{k}_{t-1}^y + s_o \hat{o}_{t-1}^y + s_y^o \hat{y}_t^o) \quad (63)$$

where $\hat{l}_t^y$, $\hat{k}_{t-1}^y$, and $\hat{o}_{t-1}^y$ represent (log) deviations of labor, capital, and oil efficiency used in the production of the final good from their steady state values, respectively, s_l is the share of labor in production of the final good, s_k is the share of capital in the final good. Note that oil usage and, hence oil prices, do not have a direct effect on real GDP; oil price movements only have an effect through labor, capital, oil efficiency, or through domestic oil production.

Given that, in the short-run, capital and efficiency are fixed (or in the medium term are subject to substantial adjustment costs), an increase in oil prices can affect output only through their effect on labor input. Thus, the responsiveness real GDP to oil price increases depends largely on the responsiveness of labor input and the elasticity of domestic oil supply. An increase in oil prices and the resulting decline oil input usage might cause a decline in labor demand, but the negative income effect (given that the United States is a net importer of oil) would lead to an increase in labor supplied. These two conflicting effects tend to mute the response of labor quantities and, hence, real GDP. Real wage rigidities would tend to lead to larger labor quantity adjustment while adjustment costs in moving labor across sectors would tend to dampen labor response. For estimating the model over our sample period, the overall response of hours worked to supply shocks is relatively small (see the above impulse response functions), suggesting substantial flexibility in how agents respond to ROW oil supply shocks. As a result, the real GDP response implied by the estimated model is relatively modest.

Why is the GDP/oil price elasticity so modest relative to the earlier, empirical literature? The general equilibrium approach taken by this paper implies that all prices respond—not just the price of oil—when there is an oil supply disruption. The price responses throughout the model generally lower the magnitude of quantity responses (for variables such as non-oil goods and of hours worked) than would be the case if prices and wages did not change. The larger price responses and lower quantity responses tend to reduce the elasticity of real GDP with respect to real oil price changes.

Despite the wage and price stickiness and the various types of adjustment costs, when taken to the data the model still finds substantially flexibility for economic agents to adjust to oil price changes. In addition, the older, purely empirical literature which was essentially a partial equilibrium or reduced form approach may not have correctly identified the effects of oil supply shocks (see Barsky and Kilian, 2004). Furthermore, the reduced form empirical literature, such as Herrera and Pesavento (2009), Kilian (2009), Kilian and Murphy (2012) and Herrera (2016), has generally found a declining importance of oil price changes on economic activity over time. Indeed, Blanchard and Gali (2010) argue that a declining oil-to-GDP ratio, increased labor market flexibility and better monetary policy have all contributed to the declining importance of oil price changes in macroeconomic fluctuations.

6. Scenario analysis

In the previous section, we used the historical data to set the values of key steady state relationships. In this section, we examine some counterfactual experiments that correspond to structural changes in the manner and the extent to which oil is used in the United States.

Specifically, we consider three alternative counterfactual scenarios. Table 6 lists various important steady state ratios for the baseline as well as for the three counterfactual scenarios.¹²

In the first scenario, we consider the effect of a substantial decline in U.S. oil consumption. In particular, we consider the case where intensity of oil usage falls substantially, lowering the percent of oil expenditures in GDP by roughly 20–25% and the percent of oil expenditures in aggregate consumption by over 50% (see Table 6).¹³ In this case the value of oil imports relative to GDP falls from 2.5% to 1.7%.

In the second scenario, we consider a substantial increase in U.S. domestic oil production—a 75% increase in domestic oil production, while the value of oil production relative to GDP increases by roughly 75% as well.¹⁴ We find the percent of oil expenditures in GDP and in consumption are relatively unchanged. Nonetheless, the ratio of oil imports to total U.S. oil consumption and the value of oil imports relative to real GDP are substantially different from the baseline scenario.

In the third scenario, we examine a combination of declining oil usage and increased domestic oil production. In this scenario, the share of oil expenditures in GDP and in consumption is substantially less than found with the baseline model (20–25% and 50%, respectively). Domestic oil production is substantially higher than in the baseline model both in terms of quantity (75%) and in value relative to GDP (67%). Oil imports are substantially lower.

Table 7 displays the implied GDP/real oil price elasticities for the baseline and three counterfactual scenarios. Comparing the scenarios suggests that reduced oil usage/intensity has more of an effect on the GDP/real oil price elasticity than does increased domestic production. Scenarios 2 and 4 which are based on large decreases in oil usage imply GDP/real oil price elasticities pretty close to zero. On the other hand, increased domestic oil production results in relatively small changes in the GDP/oil price elasticity. These findings suggest that declines in oil consumption rather than increases in domestic oil production will have more impact in lowering the sensitivity of the U.S. economy to oil price increases due to foreign supply shocks.

7. Concluding remarks and policy implications

We use a medium-sized DSGE model of the U.S. economy that also represents the world oil market, the international economy and U.S. trade to examine the effect of oil shocks on aggregate U.S. economic activity using quarterly data from 1991 through 2015. The model's parameters are established through a combination of calibration and Bayesian estimation. The estimated parameters include those over which we have informed priors and those which we have relatively uniformed priors. With this approach, most of the model's “macro” parameters are well within the range implied by typical macro models.

As expected, the impulse response functions show normal price responses to changes in oil demand or supply. Increased ROW oil demand increases world oil prices.¹⁵ Increased U.S. or ROW oil supply decreases world oil prices.

Consistent with previous research, U.S. GDP is negatively affected

¹² Because the steady state values interact with estimated parameters, it is possible for Blanchard–Kahn conditions for a unique, stable rational expectations equilibrium to hold for the baseline model but not hold for the scenario model. In this case, we restricted the comparison to parameter draws from the posterior distribution where the Blanchard–Kahn conditions held for all four models. This did not appreciably affect the posterior distribution for the GDP oil price elasticity for the baseline model.

¹³ We accomplish this by lowering the steady state level of oil intensity in the production of capital services for consumption, transportation, and goods.

¹⁴ Within the context of the model, we set the steady state total factor productivity in U.S. oil production so that in the steady state U.S. oil production is 75% larger than in the benchmark model.

¹⁵ The U.S. portion of the model is too complex to identify pure U.S. oil demand shocks. The shocks invariably have both demand and supply effects on the world oil market.

Table 6

Steady state ratios for alternative counterfactual scenarios (based on median of posterior distribution).

	Baseline	Scenario 1: Reduced oil usage intensity	Scenario 2: Increased domestic oil production	Scenario 3: Reduced oil intensity and increased domestic production
US oil production (Baseline = 1)	1.0	1.0	1.75	1.75
US oil expenditures/GDP (%)	3.7	2.9	3.8	2.9
value of US oil production/GDP (%)	1.2	1.2	2.1	2.0
value of US oil imports/GDP (%)	2.5	1.7	1.7	1.0
US oil expenditures in consumption/total US consumption (%)	3.1	1.4	3.2	1.5
quantity of US oil imports/total US oil consumption (%)	66.7	59.3	45.3	33.2
US share in world oil production (%)	7.9	8.1	13.4	13.7

Table 7

GDP/oil price elasticities for alternative counterfactual scenarios (based on posterior distribution of estimated parameters).

	Mode	Median	Mean	5th	95th
Baseline	−0.020	−0.015	−0.017	−0.031	−0.002
Scenario 1: Reduced oil usage intensity	−0.007	−0.001	−0.002	−0.010	0.009
Scenario 2: Increased domestic oil production	−0.017	−0.013	−0.015	−0.030	0.001
Scenario 3: Reduced oil usage and increased domestic oil production	−0.004	−0.001	0.000	−0.012	0.009

by reductions in ROW oil production. Following such a negative oil supply shock, ROW oil output show signs of correction, which suggests that rising prices stimulate a partial recovery in ROW production (possibly in areas of the world not affected by the disruption). U.S. oil supply rises in response to increases in oil prices brought about by the decline in ROW supply.

The historical decompositions obtained with the model show that ROW oil demand shocks—either oil specific demand shocks or shocks that affect ROW oil demand through ROW economic activity—account for most of the fluctuation in real world oil prices over the period from 1991 through 2015. ROW oil production is also an important source of the fluctuations in world oil prices, and U.S. oil production is a lesser source. In contrast, ROW oil supply shocks do not appear to have been a major contributor to U.S. real GDP fluctuations over our sample period.

Taking into account the changes in oil prices and U.S. GDP that result from a ROW oil supply shock, we find that the baseline model implies an elasticity of U.S. real GDP with respect to oil prices of around −0.015. This value would be among the less elastic estimates in the older literature, but it is consistent with estimates in the newer literature. On the other hand, we show that if long-run U.S. oil consumption falls, the elasticity of GDP with respect to oil prices falls (in absolute value) as well. In fact, exercises with the model suggest that a reduction of oil consumption relative to GDP of roughly 25% (from 3.7% to 2.9%) will push the elasticity of real GDP with respect to oil prices closer to zero.

These relatively inelastic estimates suggest that U.S. real GDP is less sensitive to oil price shocks, and oil security has become less of a concern. Moreover, scenario analysis reveals that reduced oil consumption is more effective in reducing the sensitivity of the U.S. economy to oil supply disruptions than increased domestic oil production.

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